

Multi-Modal Document Intelligence System

Assessment Project

Programming For AI

Course Assessment

Python Programming , NLTK , Image Processing , PDF Manipulation

Deadline: 07 Feb 2026

January 23, 2026

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1 Project Overview

In this comprehensive assessment project, you will build a **Multi-Modal Document Intelligence System** that demonstrates your mastery of Python fundamentals, natural language processing, image manipulation, and PDF handling. This system will process academic research papers (PDFs) to extract, analyze, and transform both textual and visual content.

1.1 Important Notice

NO AI MODELS OR AI-POWERED TOOLS ARE ALLOWED in this project. You must implement all functionality using traditional programming techniques, libraries, and algorithms. Use of ChatGPT, Claude, Copilot, or any other AI code generation/completion tools is strictly prohibited and will result in designation penalties.

2 Project Objectives

By completing this project, you will demonstrate proficiency in:

- Python programming fundamentals (data structures, functions, file I/O, error handling)
- PDF manipulation and content extraction
- Image processing and computer vision techniques
- Natural language processing and text analysis using NLTK
- Integration of multiple libraries into a cohesive application

3 Project Requirements

3.1 Part 1: PDF Processing & Page Extraction (25 points)

Task: Build a PDF processor that extracts specific pages and content.

Requirements:

1. Accept a PDF file as input (a research paper will be provided)
2. Extract pages 1, 2, 3, 4, 5, 7, 10, 15, 16, 17, 18, and 19 from the PDF
3. Save each extracted page as a separate PDF file
4. Extract all images from page 10, 15, 16, 17, 18, 19 and save them as individual image files
5. Generate a metadata report containing:
 - Total number of pages
 - Number of images found
 - File size information

Suggested Libraries: PyPDF2, pdf2image, or pdfplumber

3.2 Part 2: Image Processing & Enhancement (30 points)

Task: Apply various image processing techniques to extracted images.

Requirements:

For each image extracted from the PDF, perform the following operations:

3.2.1 Basic Transformations:

- Convert to grayscale
- Resize to 800x600 pixels (maintaining aspect ratio)
- Rotate by 45 degrees

3.2.2 Image Enhancement:

- Apply histogram equalization to improve contrast
- Apply Gaussian blur with kernel size 5x5
- Detect edges using Canny edge detection

3.2.3 Advanced Processing:

- Apply at least one morphological operation (erosion, dilation, opening, or closing)
- Create a thumbnail version (150x150 pixels)

3.2.4 Output:

- Save all processed versions with descriptive filenames
- Create a side-by-side comparison image showing original vs processed versions

Suggested Libraries: OpenCV (cv2), Pillow (PIL), NumPy

3.3 Part 3: Natural Language Processing & Text Analysis (30 points)

Task: Extract and analyze text content from the PDF using NLTK.

Requirements:

3.3.1 Text Extraction:

- Extract all text from pages 1-4 of the original PDF
- Clean the text (remove extra whitespace, special characters)

3.3.2 Tokenization & Basic Analysis:

- Tokenize text into sentences and words
- Remove stopwords
- Perform lemmatization or stemming

3.3.3 Statistical Analysis:

Calculate and display:

- Total word count
- Unique word count
- Top 20 most frequent words (excluding stopwords)
- Average sentence length

3.3.4 Advanced NLP:

- Perform Part-of-Speech (POS) tagging on the first 3 sentences of first 4 pages
- Extract all named entities (persons, organizations, locations)
- Generate word frequency distribution and visualize it using matplotlib

3.3.5 Text Visualization:

- Create a word cloud from the most frequent terms
- Generate a summary of the key topics (using frequency analysis)

Suggested Libraries: NLTK, matplotlib, wordcloud

3.4 Part 4: Integration & Presentation (15 points)

Task: Create a unified application with proper structure and documentation.

Requirements:

3.4.1 Code Organization:

- Organize code into logical functions/modules
- Use proper Python naming conventions
- Implement error handling for file operations and processing steps

3.4.2 Command-Line Interface:

- Create a main script that ties all components together
- Accept command-line arguments for input PDF path and output directory
- Display progress messages during processing (You can use tqdm library)

3.4.3 Documentation:

Include a README.md file with:

- Installation instructions
- Required dependencies (requirements.txt)
- Usage examples

- Sample output descriptions
- Add docstrings to all functions
- Include inline comments for complex operations

3.4.4 Output Report:

Generate a final text report summarizing:

- All processing steps completed
- Statistics from NLP analysis
- List of all output files created
- Processing time for each major component

4 Deliverables

Submit a ZIP file containing:

1. Source Code:

- All Python scripts (.py files)
- requirements.txt with all dependencies

2. Documentation:

- README.md file
- Any additional documentation

3. Output Folder:

- All extracted PDFs
- All processed images
- NLP analysis results (text file and visualizations)
- Final summary report

5 Evaluation Criteria

Component	Points	Criteria
PDF Processing	25	Correct extraction, metadata accuracy, error handling
Image Processing	30	All transformations applied correctly, output quality
NLP Analysis	30	Accurate text processing, meaningful statistics, proper NLTK usage
Integration	10	Code organization, CLI functionality, proper structure
Documentation	5	Clear README, docstrings, comments
Total	100	

Table 1: Project Evaluation Breakdown

6 Technical Specifications

6.1 Minimum Requirements:

- Python 3.8 or higher
- Required libraries: PyPDF2/pdfplumber, OpenCV, Pillow, NLTK, NumPy, matplotlib, word-cloud

7 Submission Guidelines

- **Deadline:** 07 Feb 2026
- **Format:** Single ZIP file named `StudentName_AIProject.zip`
- **Submission Platform:** Github
- **Academic Integrity:** This is an individual project. Collaboration on concepts is allowed, but code must be your own. **Use of AI tools is strictly prohibited.**

8 Getting Started Tips

1. Install all required libraries first: `pip install -r requirements.txt`
2. Test each component separately before integration
3. Use small sample PDFs while developing
4. Start with basic functionality, then add enhancements
5. Test error handling with invalid inputs
6. Keep backups of your work regularly

9 Resources

- Python Documentation: <https://docs.python.org/>
- NLTK Documentation: <https://www.nltk.org/>
- OpenCV Python Tutorials: <https://docs.opencv.org/>
- PyPDF2 Documentation: <https://pypdf2.readthedocs.io/>

Good luck with your project!

This is your opportunity to demonstrate comprehensive Python skills for AI applications and to get a Promotion.